Project Report

MEEG 54403: Machine learning

Uncovering Uncertainty in Student-Instructor Training Flights

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1. Introduction

1.1 Problem Definition and Relevance

General Aviation (GA) accounts for most U.S. civil aviation accidents, with a fatal accident rate of about 0.76 accidents per 100,000 flight hours in 2023 [1]. Past analyses show that 70-80 % of aviation accidents are attributable to human error rather than mechanical issues [2, 3]. Training operations represent roughly 14 % of GA accidents, indicating that student–instructor flights remain a meaningful source of human-error events [4, 5]. These flights are particularly vulnerable to uncertainty-driven factors such as hesitation, miscommunication, confusion, degraded situational awareness, and delayed decision-making patterns frequently reflected in safety reports.

NASA's Aviation Safety Reporting System (ASRS) [6] offers thousands of real-world safety narratives, but most of them exist as long, unstructured text reports. Traditional human-factors analysis requires experts to manually read each narrative, as a result, many recurring safety patterns remain hidden.

This project applies unsupervised text clustering to ASRS training-flight reports to discover the major uncertainty patterns present in student–instructor events. Since ASRS narratives lack predefined labels, unsupervised learning is well-suited for identifying natural groupings such as communication breakdowns, traffic or workload misjudgments, and procedural deviations. By embedding both pilot-written narratives and expert-written synopses, the analysis compares how these two text sources cluster and then derives thematic categories of uncertainty from the resulting groups.

Automating this analysis through machine learning provides two benefits for the aviation industry. Methodologically, it demonstrates a scalable workflow for processing large bodies of safety text, reducing reliance on labor-intensive manual reviews. From a human-factors perspective, the resulting clusters highlight recurring uncertainty themes that can inform curriculum design, scenario-based training, and targeted intervention strategies for student pilots and instructors. Ultimately, this approach supports more data-driven improvements in flight-training safety.

1.2 Objectives

The objectives of the project are:

1. To discover the dominant uncertainty themes that appear in student instructor ASRS reports.
2. To cluster narrative texts using machine learning and interpret the resulting patterns.
3. To compare the narrative clusters with clusters derived from the synopses to evaluate thematic consistency.
4. To understand what the identified themes reveal about the nature of uncertainty in training flights.

2. Dataset

ASRS is a nationwide voluntary reporting system where pilots, student pilots, flight instructors, and other aviation personnel document safety incidents, making it a valuable resource for human-factors research. This project uses NASA's ASRS data, focusing on GA events involving student–instructor training flights from 2020 to 2024. After filtering, the dataset contained 1,451 reports detailing sequences of events from actual training flights.

Each ASRS report includes two key text components:

- **Pilot Narrative:** a detailed, free-form description written by the reporting pilot or instructor.
- **Analyst Synopsis:** a concise summary written by an ASRS analyst that highlights the core issue of the event [7].

This dual structure allows the study to examine how uncertainty is expressed both from the pilot's perspective (rich, experiential detail) and from an expert perspective (standardized interpretation).

2.1 Dataset Characteristics

After filtering for student-instructor training flight, the dataset captures a wide range of uncertainty-related scenarios common in GA operations, including landing or pattern-entry deviations, miscommunication with ATC or other aircraft, visibility challenges, and maneuver-execution errors.

Narrative Length Distribution:

- Mean length: 250.58 words

- Standard deviation: 196.39 words
- Minimum: 14 words
- Maximum: 2215 words

This large variation between the maximum and the minimum length distributions reflects the highly unstructured nature of pilot reporting. Very short narratives provide limited contextual information, while very long ones may introduce noise. Both extremes influence the separation of clusters because unsupervised learning relies on semantic consistency across samples.

Synopsis Length Distribution:

- Mean length: 23.97 words
- Standard deviation: 10.66 words
- Minimum: 5 words
- Maximum: 86 words

Analyst-written synopses are more uniform ranging from 5 to 86 words, which makes them useful as an external reference when validating the narrative-based cluster patterns.

Below are sample excerpts:

Narrative Example:

"While practicing commercial maneuvers in the practice area during the initial climb of a lazy eight; I observed a tailwheel aircraft approximately 100 ft vertically and 100 ft horizontally from our aircraft. I immediately made an evasive climbing maneuver to avoid the aircraft. The aircraft was not observed on ADS-B and did not make radio calls for the practice area.

Synopsis Example:

"C172 pilot reported an NMAC in the practice area."

These examples illustrate how narratives capture high-detail situational context, whereas synopses compress the event into its essential safety-relevant theme.

2.2 Keyword Filtering Assumptions and Limitations

Reports were selected using keywords such as "student," "instructor," and "training flight." This method assumes that reporters explicitly identify the training context. As a result:

- Some relevant training events may be excluded if the narrative does not contain the expected keywords.
- Some non-training reports may be included if they mention these keywords in unrelated contexts.

This introduces noise into the dataset, which can influence cluster purity and should be considered when interpreting results.

2.3 Pre-processing

To prepare the text for machine learning, the following preprocessing steps were applied:

1. **Data Filtering:** Retained only GA reports that matched training-related keywords.
2. **Text Cleaning:** Removal of empty narratives, lowercasing, stripping punctuation, and removing stray filler characters.

2.4 Baseline Comparison

A manual-reading approach was used as the baseline. Traditionally, experts review ASRS narratives one by one and group them into themes using their own judgment. This method is familiar in human-factors work but has clear limitations: it is slow, cannot handle large datasets efficiently, and often leads to inconsistent results across reviewers. It also makes it hard to notice deeper patterns that only emerge when many reports are analyzed together. In contrast, the machine learning model groups all 1451 narratives based on semantic similarity at once, providing a faster, more consistent, and more scalable way to identify uncertainty patterns.

3. Methodology

This section describes the workflow used to process the ASRS reports, generate semantic embeddings, evaluate potential cluster structures, and identify uncertainty themes in student–instructor training events and clustering using KMeans [8].

3.1 Text Embedding Using Sentence Transformers

Since clustering algorithms cannot process raw text, each narrative and synopsis was converted into a numerical embedding [9] using the SentenceTransformer model all-MiniLM-L6-v2 [10]. This embedding model generates dense vector representations that capture semantic similarity, allowing reports that describe similar uncertainty events to be grouped together even if the wording differs. The embedding step transforms unstructured text into a machine-readable format and serves as the feature space for all clustering experiments.

3.2 Exploratory Cluster Analysis

Before selecting a final clustering model, several internal evaluation metrics were computed for values of k ranging from 2 to 15 as shown in figure 1. These metrics help determine how many clusters best reflect the structure of the data:

- **Elbow Method**: analyzes the rate of decrease in within-cluster inertia [11].
- **Silhouette Score**: measures how well-separated the clusters are [12].
- **Calinski–Harabasz Score**: evaluates compactness and separation [13].
- **Davies–Bouldin Index**: measures average cluster similarity where lower is better [14].

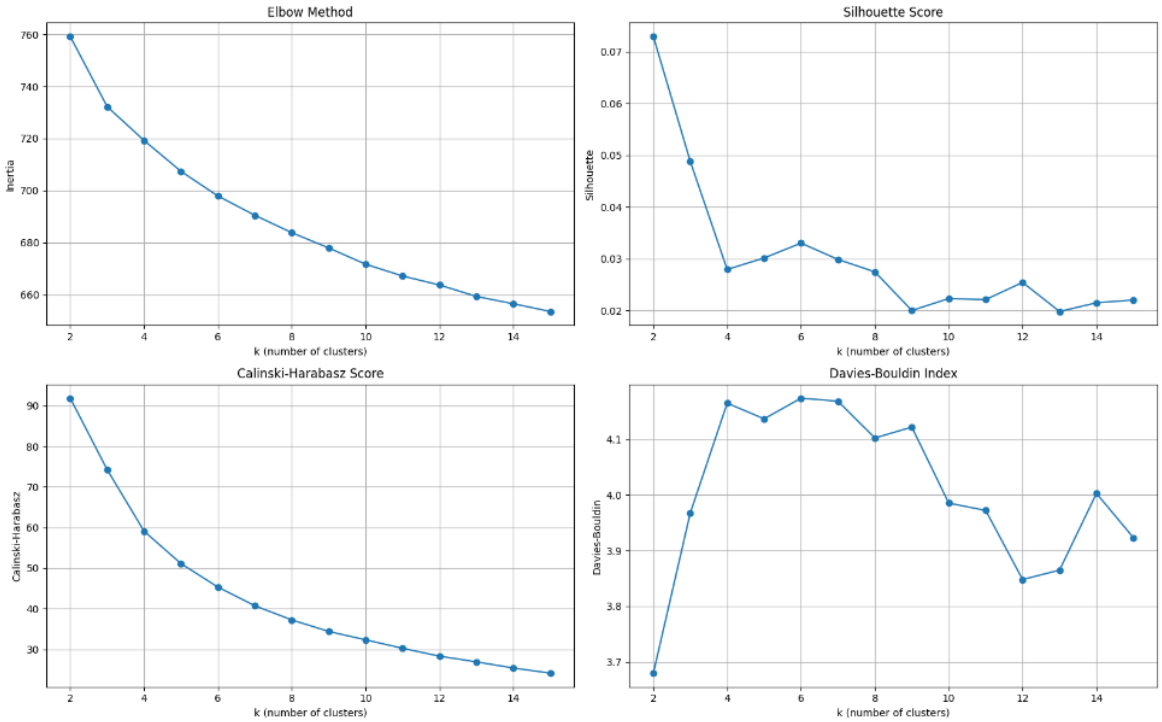

Figure 1: Internal clustering evaluation metrics on Narratives Texts

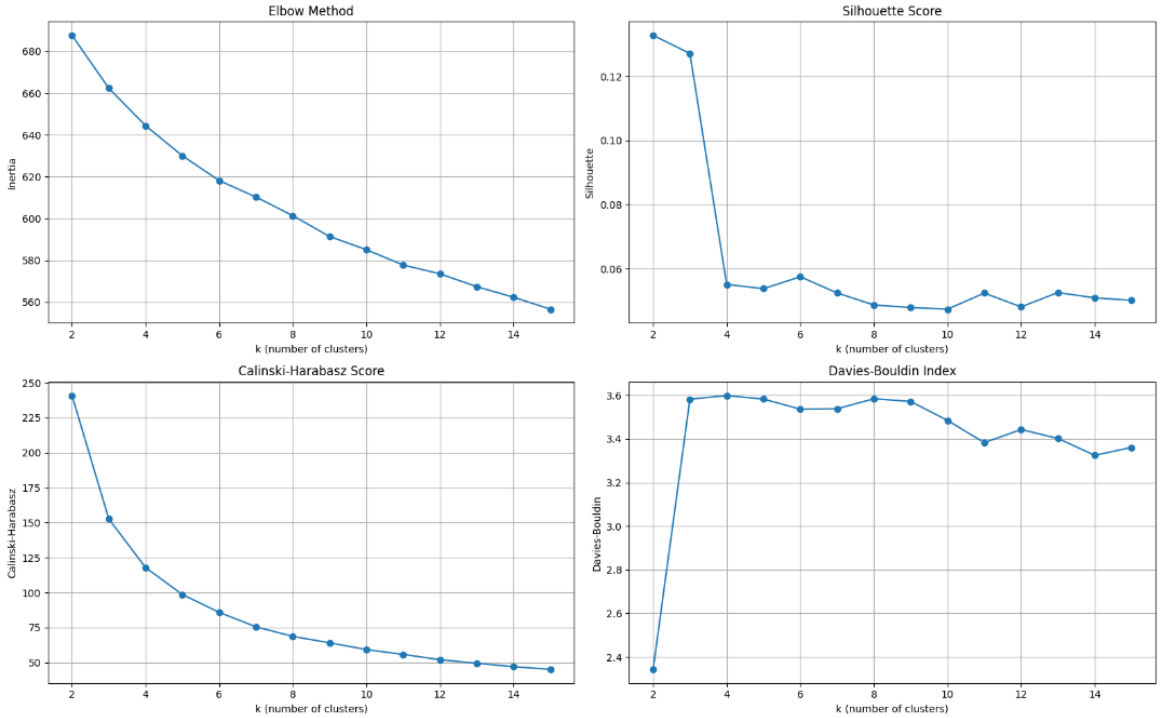

Figure 2: Internal clustering evaluation metrics on Synopsis Texts

Across all four metrics, k = 2 consistently emerged as the most appropriate choice. The Silhouette Score reached its highest value at k = 2, indicating the clearest separation at this cluster count. The Davies–Bouldin Index reached its lowest value at k = 2, suggesting optimal compactness. The Calinski–Harabasz Score was highest at k = 2, showing the strongest cluster definition. The Elbow Method indicated that most of the improvement occurs at k = 2, with diminishing returns afterward.

These results show that the dataset naturally separates into two broad semantic groups, which is typical for subjective human-written safety narratives. Importantly, the same internal evaluation metrics were also computed on the synopsis embeddings as shown in figure 2, and they likewise indicated that k = 2 was the most appropriate number of clusters. This agreement between the narratives and synopses provides additional validation for selecting k = 2 as the final cluster count. Because all internal metrics pointed to the same structure across both text forms, k = 2 was chosen for the final clustering stage.

3.3 Domain Justification of Clusters

The Human Factors Analysis and Classification System (HFACS), a framework developed by Wiegmann and Shappell for categorizing human error in aviation accidents, identifies three types of pilot errors: decision errors, skill-based errors, and perceptual errors [15]. In training-flight narratives, perceptual errors rarely appear as a separate category. They usually fall into either a decision mistake or a handling mistake, for example, misjudging traffic spacing and then choosing the wrong action or applying incorrect control inputs. As a result, the events described in ASRS reports tend to fall into two clear human-factors patterns. The model's two clusters align directly with these two types of pilot errors making k = 2 not only statistically supported but also consistent with how safety-related errors typically manifest in GA training.

3.4 Final Clustering with KMeans

The final clustering was performed using KMeans, with parameters: k = 2, random_state = 42 (for reproducibility), n_init = 10 (for stable cluster centers). KMeans clustering was ran separately on the narrative embeddings and synopsis embeddings. This allowed comparison between the two models to examine how consistently both forms of text separated into the same two uncertainty themes.

3.5 Clustering Evaluation Criteria

To evaluate the usefulness of the clustering results, three criteria are applied:

1. **Cluster coherence**: whether reports grouped together describe similar types of events.
2. **Narrative synopsis alignment**: whether themes extracted from narratives match the themes derived from synopses, which act as a consistency check.
3. **Human-factors relevance**: whether the identified themes correspond to known categories such as HFACS decision errors, skill-based errors, or perceptual issues.

3.6 Software and Implementation

All steps were carried out in Python in google colab, using pandas for data loading and preprocessing, sentence-transformers for embedding generation. scikit-learn for clustering and internal metric evaluation, matplotlib for plotting the exploratory metric figures, umap for clustering visualization. The project ran entirely on a standard CPU environment. The workflow is fully reproducible, and a fixed embedding model (all-MiniLM-L6-v2) was used to generate deterministic vector outputs. Random seeds, such as random_state=42, were set to ensure consistent clustering results. The pipeline also follows a transparent and well-documented sequence: loading the ASRS data, cleaning the text by lowercasing and removing empty entries or stray symbols, generating semantic embeddings, evaluating potential cluster structures using silhouette, performing KMeans clustering with k = 2, comparing the narrative and synopsis results and comparing the model with others. By following this same sequence with the same dataset and libraries, the analysis can be replicated exactly.

4. Results and Analysis

This section presents the cluster alignment results, visualizations, cluster interpretations, aviation insight and model justification.

4.1 Narrative and Synopsis UMAP Visualizations

Uniform Manifold Approximation and Projection (UMAP) was used to visualize how the high-dimensional embeddings group in two dimensions.

a. **Narrative UMAP:** the narrative plot as shown in figure 3 shows two broad clusters with partial overlap. This is expected because narrative descriptions vary widely in length and detail, and humans often write in less structured ways.

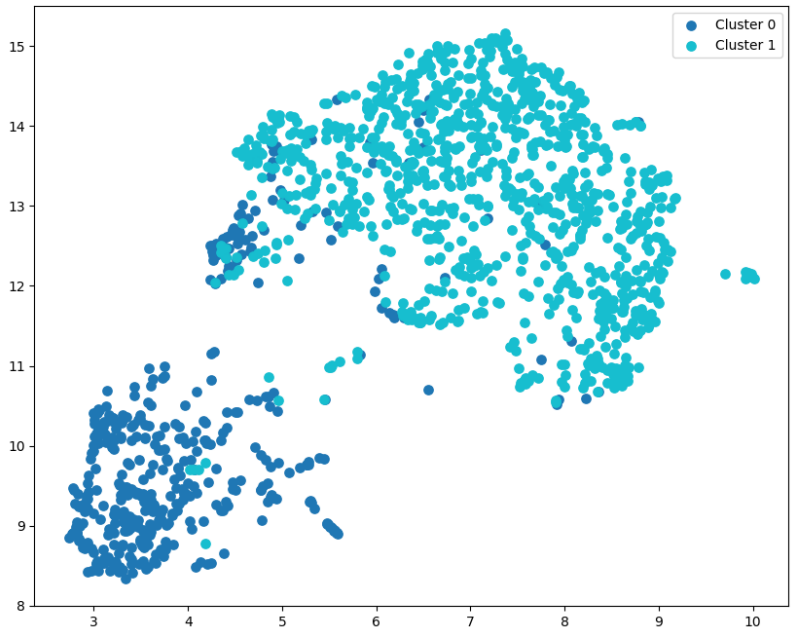

Figure 3: UMAP Visualization of ASRS Narratives Clusters

b. **Synopsis UMAP:** the synopsis clusters shown in figure 4 show clearer separation, consistent with their standardized structure.

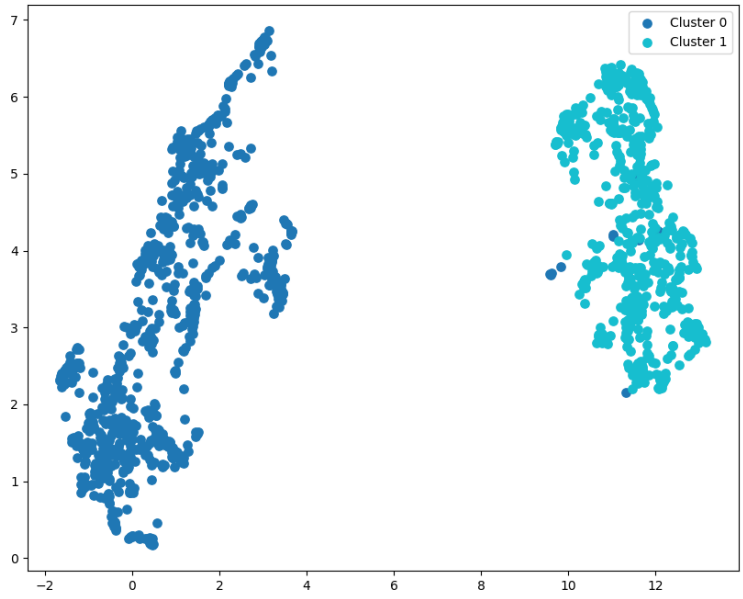

Figure 4: UMAP Visualization of ASRS Synopsis Clusters

4.2 Cluster Alignment Between Narratives and Synopses

A confusion matrix shown in figure 5 was used to evaluate how well the narrative clusters aligned with the clusters independently generated from the synopsis embeddings. Because synopses are shorter, more structured, and written by ASRS analysts, they serve as a reasonable pseudo-ground truth reference.

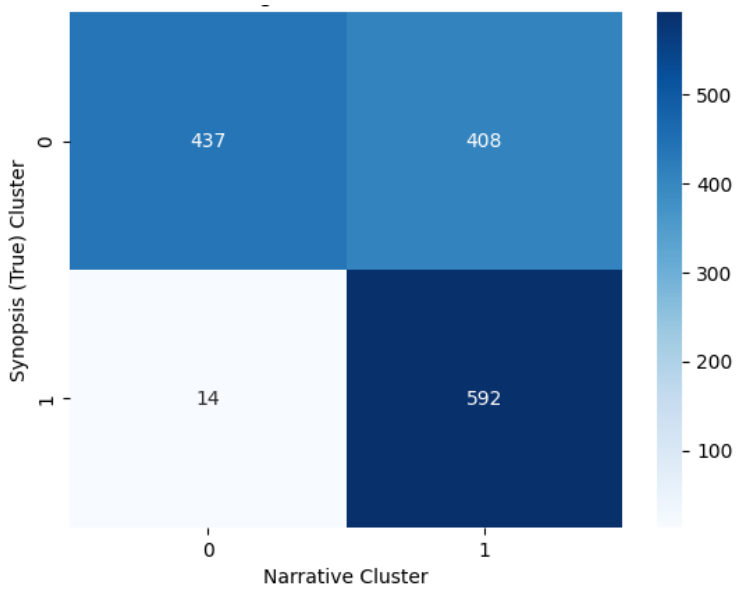

Figure 5: Cluster Alignment Confusion Matrix

4.3 Cluster Interpretability Through Word Clouds

Word clouds were generated to identify the dominant vocabulary within each cluster [16].

a. **Cluster 0 (Execution Uncertainty):** Common terms as shown in figure 6 include landing, control, engine, left, back, taxiway, brake, rudder, rollout, gear

Figure 6: Cluster 0 Word Cloud

This cluster contains reports describing aircraft handling and execution errors, including landing misjudgment, incorrect control inputs, configuration mistakes (e.g., gear, flaps, throttle), runway and taxiway deviations, student mistakes during physical maneuver execution

Sample Narrative:

"After making a safe landing on runway; the student accidentally released the 'Johnson Bar' gear handle causing the gear to retract on the landing roll. I immediately tried to force the handle back into its gear down and locked position; while the belly of the aircraft impacted the surface of the runway and the aircraft began to depart the runway to the right. I immediately pulled the mixture into the idle cutoff; and steered the aircraft using the rudder controls to avoid runway lights. The aircraft came to a rest on the right side of the runway in the grass. I turned off the master; fuel valve; and took the key out of the ignition. We both left the aircraft immediately with no injuries to either of us at all."

This confirms that Cluster 0 represents execution uncertainty.

b. Cluster 1 (Information Uncertainty): Common terms as shown in figure 7 include traffic, pattern, call, downwind, approach, radio, altitude, collision, position

Figure 7: Cluster 1 Word Cloud

Reports in this cluster involve: miscommunication or missed radio calls, pattern-entry conflicts, traffic awareness failures, incorrect assumptions about other aircraft positions, incomplete or inaccurate situational awareness

Sample Narrative:

"I was in the traffic pattern in a C172 with a student pilot; teaching pattern work. I announced 45 degree entry to right downwind Runway XX at ZZZ. When we reached the downwind leg; I announced downwind leg. I was startled to hear from Aircraft Y that he too was downwind Runway XX. I immediately asked him to state your position. He said he was over the freeway. The freeway is about 3 miles from the runway. I told him that the freeway was not in the pattern. Moments later; once abeam the numbers to Runway XX; I was startled to see Aircraft Y go directly underneath us toward the runway; then make a 360 and join the pattern below us. I asked him: 'Aircraft Y; what are you doing?' He said; 'I saw you in the pattern so I decided to go below you.' This is highly unsafe practice. He nearly collided with us. Later; he landed on Runway XX while

we were still on the runway in our landing roll. Someone should reach out to this pilot and provide remedial training. A FAA report has been filed."

This cluster clearly reflects information uncertainty, issues arising from ambiguity, incomplete information, or misunderstanding between pilots.

4.4 Cluster Alignment to Aviation

The two clusters identified in this study match well-known human-factors categories in aviation. Cluster 0 (Execution Uncertainty) reflects HFACS Skill-Based Errors, where mistakes arise from control inputs, landing technique, configuration handling, or other aircraft-manipulation tasks. Cluster 1 (Information Uncertainty) aligns with HFACS Decision Errors, including miscommunication, traffic-pattern misunderstandings, and loss of situational awareness. This shows that the model's two clusters capture the main types of uncertainty that commonly affect student pilots during training flights.

4.5 Model Comparison

To evaluate model performance, three approaches were compared: TF-IDF, SBERT without text cleaning, and SBERT with text cleaning. The differences were clear. TF-IDF produced a very low silhouette score (0.0137), indicating that its clusters had almost no real structure. In contrast, SBERT embeddings created much stronger separation, with silhouette scores around 0.072 for the uncleaned text and 0.073 after cleaning. The UMAP visualizations reinforced these results: TF-IDF clusters were heavily mixed and poorly defined, while SBERT showed two distinct regions, and the separation became even clearer after basic text cleaning. These findings show that SBERT captures the semantic meaning of the narratives far better than TF-IDF, which relies only on word frequency. Text cleaning also helps by reducing noise and making the embedding space more consistent. Overall, SBERT with text cleaning is the most effective and stable model for identifying meaningful clusters in ASRS training-flight narratives.

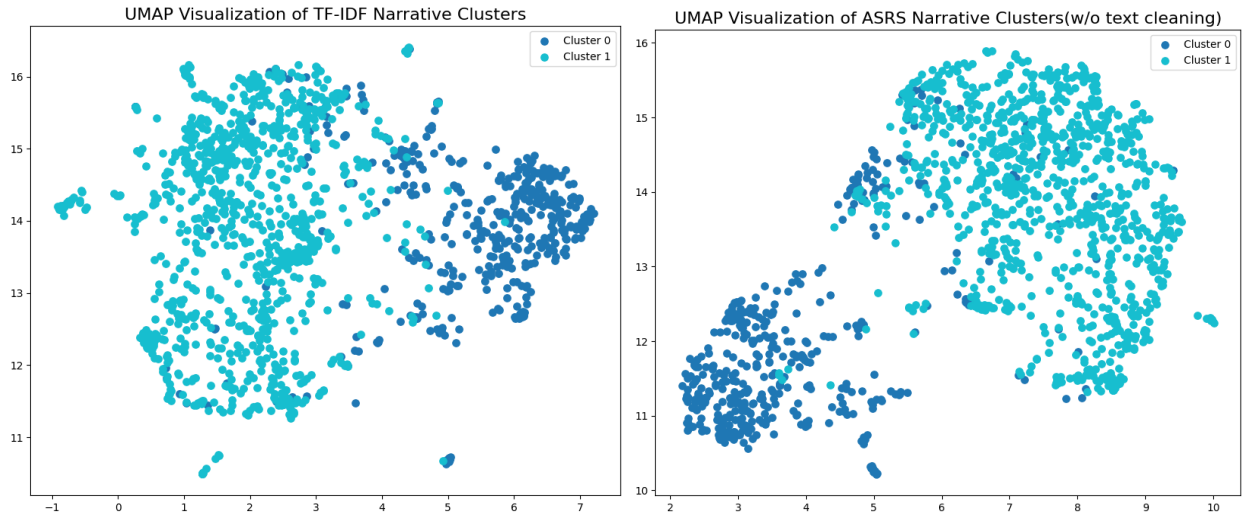

Figure 8: Comparison of clustering methods using UMAP visualization. (a) TF-IDF (b) SBERT (no cleaning)

5. Discussion

The clustering results show that student instructor training reports naturally separate into two main uncertainty themes: Information Uncertainty and Execution Uncertainty. Information Uncertainty makes up about 69 percent (1000 out of 1451) of all events, suggesting that most training issues stem from miscommunication, unclear intentions, traffic misjudgment, and reduced situational awareness rather than from physical aircraft handling mistakes. This aligns with GA training environments, where students are still learning radio skills, pattern awareness, and how to manage rapidly changing traffic.

Execution Uncertainty, representing about 31 percent (451 out of 1451) of events, captures errors in landing technique, configuration changes, or control inputs. These incidents occur during mechanically demanding phases of flight and highlight the need for continued emphasis on basic handling skills, especially during landing and rollout.

The consistent cluster separation across both narratives and synopsis reinforces that these two themes are meaningful and stable. From a training and engineering perspective, these findings suggest that improving communication training, traffic-awareness exercises, and situational-awareness instruction may have a greater impact on reducing incidents than focusing solely on maneuver execution. The analysis also shows that machine learning can effectively extract useful

safety patterns from unstructured GA text, offering a faster and more practical tool for identifying where student pilots struggle most.

6. Limitations

While the clustering results provide useful insights, a few factors limit how they should be interpreted. ASRS narratives differ greatly in length and detail, which affects how consistently the model can group them. The reporting system is also voluntary, so the data reflects only incidents that pilots decide to report, creating some reporting bias [17]. Finally, the analysis uses only text and does not include flight data such as altitude or airspeed, meaning the model can reveal general themes but not detailed performance differences.

7. Conclusion

This project used unsupervised machine learning to analyze ASRS reports from GA training flights and identify the main types of uncertainty student pilots experience. By converting narratives and synopses into semantic embeddings and clustering them with KMeans, the analysis revealed two consistent themes: Information Uncertainty and Execution Uncertainty. Information-related issues such as miscommunication, unclear intentions, and traffic-awareness problems accounted for most events, while execution errors made up a smaller portion.

These findings show that many training incidents arise from how information is interpreted and shared rather than from mechanical handling alone. This highlights opportunities for improving communication skills, pattern-entry procedures, and situational-awareness training in GA environments. The ability of machine learning to uncover these patterns also demonstrates the value of text-based safety data as a tool for identifying training weaknesses and supporting human-factors engineering efforts.

Overall, the study provides a clearer understanding of the uncertainty challenges faced by student pilots and shows how data-driven methods can contribute to safer and more effective training practices.

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Appendix

A.1 Notebook

All analysis for this project was completed in Google Colab using the notebook:

Toluwalase_Olatunji_Project_ML.ipynb

A.2 Dataset

The dataset used in this project was manually uploaded into the Colab session. To reproduce the results:

1. Open Google Colab.
2. Upload the notebook file.
3. Upload the dataset file ASRS_DBOnline.csv (the same file submitted with this report).
4. Run all notebook cells in order.

A.3 Code Pipeline Overview

The notebook contains the full reproducible pipeline:

1. Load and filter ASRS GA training data

2. Clean text (lowercase, remove blanks, remove stray symbols)
3. Generate narrative and synopsis embeddings
4. Compute clustering metrics for k = 2–15
5. Select k = 2 as optimal
6. Run final KMeans clustering
7. Compare narrative clusters with synopsis clusters
8. Produce UMAP visualizations and evaluation metrics
9. Model Comparison

All figures included in the report were generated directly from this notebook.

A.4 Reproducibility

The project is fully reproducible because:

- The notebook includes every step of the analysis.
- The dataset file is provided with the submission.
- Random seeds (random_state=42) ensure identical clustering results.
- The embedding model (all-MiniLM-L6-v2) generates deterministic vectors.

Anyone who uploads the notebook and dataset into Google Colab and runs all cells will obtain the same results, figures, and interpretations.